Heart Disease Prediction

Using K-Nearest Neighbors Classification

CSE475: Machine Learning

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1. Introduction

Heart disease remains one of the leading causes of mortality worldwide and continues to be a major public health concern. Early detection of heart disease is crucial for timely treatment and risk reduction. In recent years, machine learning techniques have gained significant attention in the healthcare domain due to their ability to identify complex patterns in medical data and support clinical decision-making.

This project focuses on developing a K-Nearest Neighbors (KNN) based binary classification model to predict whether a patient is likely to have heart disease. The study follows a structured machine learning workflow, including data preprocessing, exploratory data analysis, model training, and performance evaluation.

2. Dataset Description

The dataset used in this project is the Heart Disease Dataset, a widely used benchmark dataset in the field of medical machine learning. It contains clinical records of patients and is commonly used for binary classification tasks related to heart disease prediction. The dataset consists of 14 attributes, including 13 input features and 1 binary target variable indicating the presence or absence of heart disease.

2.1 Feature Summary

Feature	Type	Description
age	Continuous	Age of the patient in years
sex	Binary	1 = Male, 0 = Female
cp	Categorical	Chest pain type (0 = typical angina, 1 = atypical, 2 = non-anginal, 3 = asymptomatic)
trestbps	Continuous	Resting blood pressure (mm Hg) at hospital admission
chol	Continuous	Serum cholesterol in mg/dl
fbs	Binary	Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)
restecg	Categorical	Resting ECG results (0 = normal, 1 = ST-T abnormality, 2 = LVH)
thalach	Continuous	Maximum heart rate achieved during exercise
exang	Binary	Exercise-induced angina (1 = yes, 0 = no)
oldpeak	Continuous	ST depression induced by exercise relative to rest
slope	Categorical	Slope of the peak exercise ST segment (0-2)
ca	Categorical	Number of major vessels colored by fluoroscopy (0-4)
thal	Categorical	Thalassemia type (0 = normal, 1 = fixed defect, 2 = reversible defect)
target	Binary	Heart disease diagnosis: 1 = Disease, 0 = No Disease

2.2 Dataset Statistics

The dataset originally contained 1,025 records. After preprocessing and duplicate removal, the cleaned dataset retained 302 unique samples.

Features: 13 input features + 1 binary target

Target distribution (after deduplication): 54.6% Disease (1), 45.4% No Disease (0)

Missing values: None detected

3. Data Preprocessing

3.1 Duplicate Row Detection and Removal

A critical preprocessing finding was the presence of a large number of exact duplicate rows in the raw dataset. Upon inspection, 723 out of 1,025 rows were identified as exact duplicates. These duplicates are likely artifacts introduced during dataset augmentation or redistribution.

Retaining duplicate records can lead to data leakage, particularly in distance-based algorithms such as K-Nearest Neighbors (KNN). In such cases, the model may achieve artificially high performance (e.g., 100% accuracy for K=1) by matching identical samples in both training and testing sets. To ensure a fair evaluation, all duplicate records were removed prior to model training.

Data Quality	Count
Total raw rows	1,025
Duplicate rows removed	723
Unique samples retained	302

3.2 Missing Value Check

After removing duplicates, the remaining 302 records were examined for missing values. No null or NaN values were detected in any feature, confirming that the dataset is complete and suitable for further analysis.

3.3 Feature Scaling

K-Nearest Neighbors (KNN) is a distance-based algorithm that relies on Euclidean distance to compute similarity between data points. Therefore, features with larger numerical ranges (e.g., cholesterol (chol) or resting blood pressure (trestbps)) can disproportionately influence the distance calculation compared to smaller-scale or binary features.

To address this issue, all numerical features were standardized using StandardScaler from scikit-learn, transforming them to have zero mean and unit variance. Importantly, the scaler was fitted only on the training data and then applied to the test set to avoid data leakage.

3.4 Train-Test Split

The dataset was split into training and testing sets using an 80:20 ratio. Stratified sampling was applied to preserve the original class distribution in both subsets. A fixed random seed (random_state = 42) was used to ensure reproducibility.

Training set size: 241 samples

Test set size: 61 samples

4. Exploratory Data Analysis (EDA)

4.1 Descriptive Statistics

After deduplication, the cleaned dataset containing 302 patient samples was analyzed to understand the distributional properties of all features. Table-based summary statistics, including mean, standard deviation, minimum, median, and maximum values, were computed for each variable.

Feature	Mean	Std Dev	Min	Median	Max
age	54.4	9.1	29	56	77
trestbps	131.6	17.6	94	130	200
chol	246.7	51.8	126	241	564
thalach	149.6	22.9	71	153	202
oldpeak	1.04	1.16	0.0	0.8	6.2

4.2 Class Distribution

After removing duplicate records, the cleaned dataset contained 302 unique patient samples. The target variable exhibited a relatively balanced distribution, with 164 patients diagnosed with heart disease and 138 patients without heart disease.

Figure 1 illustrates the distribution of the target classes in the cleaned dataset. Although the heart disease class is slightly more prevalent, the difference is not substantial enough to introduce significant class imbalance or bias into the model training process.

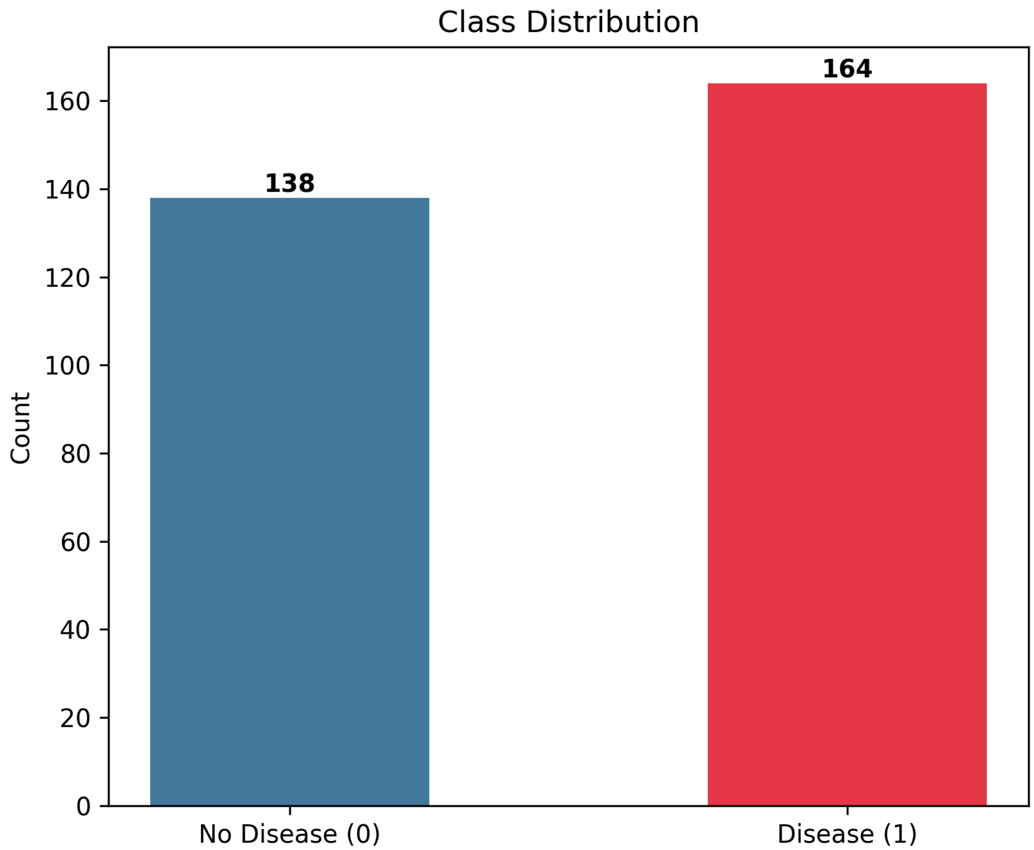

Figure 1: Class Distribution of the Cleaned Heart Disease Dataset

4.3 Correlation Analysis

A Pearson correlation matrix was computed to examine the linear relationships between the input features and the target variable (heart disease presence). This analysis helps in identifying which clinical variables are more strongly associated with heart disease and provides useful interpretability for the model.

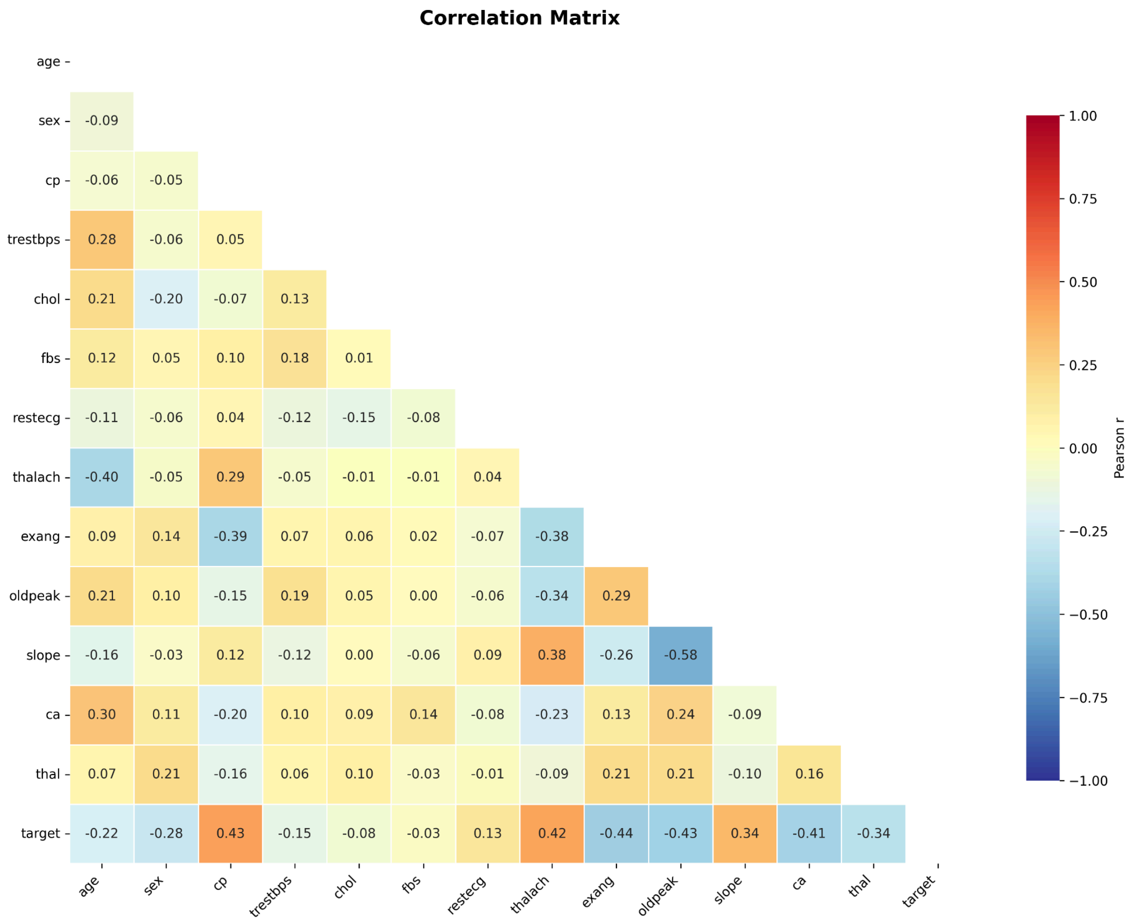

Figure 2. Pearson Correlation Matrix of Clinical Features and Heart Disease Target Variable

The top features most correlated with the target variable (heart disease presence) are listed below:

Feature	Correlation with Target (|r|)
cp (chest pain type)	0.43
thalach (max heart rate)	0.42
exang (exercise angina)	0.43
oldpeak (ST depression)	0.42
ca (vessel count)	0.45
thal (thalassemia type)	0.34

Key EDA observations:

- thalach (maximum heart rate) is negatively correlated with disease - disease patients tend to have lower max heart rate during exercise.
- cp (chest pain type) is a strong predictor; asymptomatic chest pain is associated with higher disease prevalence.
- oldpeak (ST depression) is significantly higher in disease-positive patients, reflecting cardiac stress during exercise.
- ca (number of fluoroscopy-colored vessels) shows the highest individual correlation with the target.

5. KNN Model

K-Nearest Neighbors (KNN) is a non-parametric, instance-based learning algorithm. For a given test sample, KNN computes the Euclidean distance to all training samples, selects the K closest neighbors, and classifies the test point by majority vote among those K neighbors.

KNN has no explicit training phase - the model memorizes all training data and defers computation to prediction time. This makes it sensitive to the scale of features (requiring StandardScaler) and the choice of K.

5.1 Hyperparameter Tuning - Selecting Optimal K

The hyperparameter K (number of neighbors) directly controls the bias-variance tradeoff:

- Small K (e.g., K=1): Low bias but high variance, leading to overfitting and sensitivity to noise.
- Large K: Higher bias but lower variance, resulting in overly smooth decision boundaries.
- Optimal K: Provides the best generalization performance on unseen data.

To determine the optimal value of K, 5-fold Stratified Cross-Validation was performed on the training dataset, evaluating K values from 1 to 30. The value of K that achieved the highest mean cross-validation accuracy was selected as the optimal parameter.

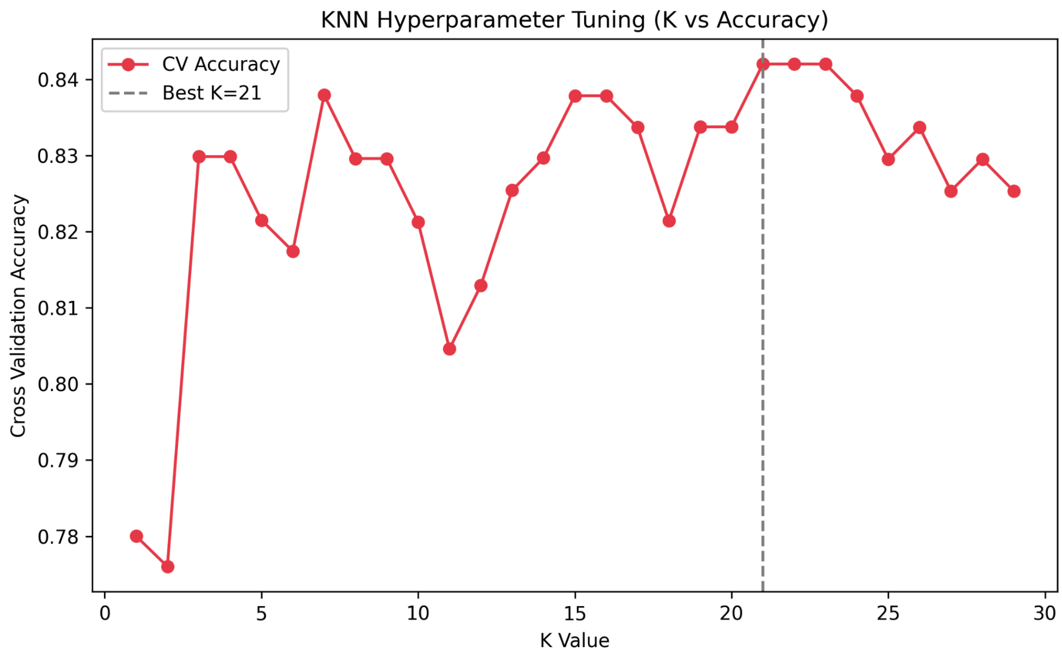

Figure 3. Cross-Validation Accuracy vs Different K Values

The best performing model was obtained with:

Optimal K found: 21

Cross-validation accuracy: 84.6%

5.2 Final Model Configuration

The final K-Nearest Neighbors (KNN) model was configured using the optimal hyperparameters selected through cross-validation. The model setup ensures consistency, reproducibility, and proper handling of feature scaling and distance-based computation.

The key configuration details of the final model are summarized below:

Parameter	Value
Algorithm	KNeighborsClassifier
n_neighbors (K)	21
Distance metric	Euclidean (default)
Weighting	Uniform (all neighbors equal weight)
Feature scaling	StandardScaler (mean=0, std=1)
CV strategy	StratifiedKFold (n_splits=5)

6. Model Performance Evaluation

6.1 Performance Metrics

After training the final K-Nearest Neighbors (KNN) model (K = 21), its performance was evaluated on the held-out test set consisting of 61 samples. The model was assessed using standard classification metrics to ensure a comprehensive evaluation.

The results are summarized below:

Metric	Value
Accuracy	85.25%
Precision	83.33%
Recall	90.91%
F1-Score	86.96%

6.2 Confusion Matrix Analysis

Figure 4 illustrates the confusion matrix of the K-Nearest Neighbors (KNN) classifier evaluated on the test dataset. It provides a comprehensive overview of the model's classification performance by showing the distribution of correct and incorrect predictions across both classes.

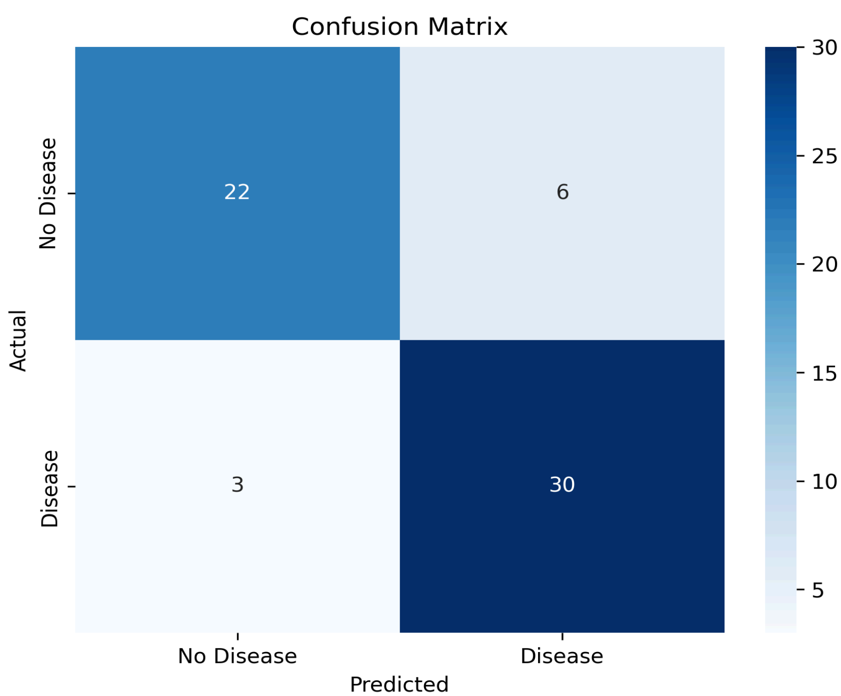

Figure 4: Confusion Matrix of the KNN Classifier

- As depicted in Figure 4, the model correctly classified 22 patients without heart disease (True Negatives) and 30 patients with heart disease (True Positives). These results indicate that the model has a strong ability to correctly identify both healthy and diseased cases.
- On the other hand, the model misclassified 6 healthy patients as having heart disease (False Positives). While this may lead to additional diagnostic tests, such errors are generally acceptable in medical applications compared to missing actual disease cases.
- Additionally, the model failed to correctly identify 3 patients with heart disease (False Negatives), classifying them as healthy. In clinical practice, False Negatives are particularly critical, as they may delay treatment and increase health risks.

7. Conclusion

This project successfully developed and evaluated a K-Nearest Neighbors (KNN) classifier for predicting heart disease using the Cleveland Heart Disease dataset. The study followed a complete machine learning pipeline, including data preprocessing, exploratory data analysis, model training, and performance evaluation.

A significant issue identified in the dataset was the presence of 723 duplicate records (70.5% of the dataset). Removing these duplicates was essential to prevent data leakage and ensure a realistic evaluation of model performance.

Exploratory data analysis revealed that features such as chest pain type (cp), maximum heart rate achieved (thalach), ST depression (oldpeak), and number of major vessels (ca) are among the most important predictors of heart disease.

Since KNN is a distance-based algorithm, feature scaling using StandardScaler played a crucial role in ensuring that all features contributed equally to the distance calculation. Additionally, 5-fold stratified cross-validation was used to select the optimal number of neighbors, with K = 21 providing the best performance.